

# CSS Complete Guide - Basics to Advanced

## Introduction to CSS

CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation of HTML documents. It allows you to apply styles such as colors, fonts, spacing, and layouts.

## CSS Syntax

A CSS rule-set consists of a selector and a declaration block. For example:

```
```\css
h1 { color: blue; font-size: 20px; }
```
```

Here, `h1` is the selector and `{ color: blue; font-size: 20px; }` is the declaration block.

## Selectors

CSS provides various types of selectors:

- Element Selector
- Class Selector (.class)
- ID Selector (#id)
- Grouping Selector
- Universal Selector
- Attribute Selector
- Pseudo-classes and Pseudo-elements

## Box Model

Every HTML element is a rectangular box and has four components:

1. Content
2. Padding
3. Border
4. Margin

Understanding the box model is crucial for layout control.

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## Positioning and Display

- `static`, `relative`, `absolute`, `fixed`, `sticky`
- `display: block`, `inline`, `inline-block`, `flex`, `grid`, `none`

These properties define how elements are laid out on the page.

## Flexbox

A modern layout model that allows responsive alignment of items. Key properties:

- display: flex
- justify-content
- align-items
- flex-direction
- flex-wrap

## CSS Grid

CSS Grid Layout provides a two-dimensional grid-based layout system. Example:

```
```css
display: grid;
grid-template-columns: repeat(3, 1fr);
```
```

## Responsive Design

Responsive design ensures that your website looks good on all devices. Use:

- Media Queries
- Relative units (% , em, rem)
- Mobile-first design

## Advanced Topics

- CSS Variables: `--main-color: #333;`
- Transitions & Animations

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- Custom Fonts using ``@font-face``
- CSS Functions like ``calc()``, ``clamp()``
- Preprocessors like SASS/SCSS

## CSS Best Practices

- Keep your CSS DRY (Don't Repeat Yourself)
- Use semantic class names
- Group related styles
- Minify CSS for production
- Use tools like Autoprefixer